INDUSTRIAL ASSIGNMENT- 2 Requirements Gathering
STUDENT INTERNSHIP AND PLACEMENT MANAGEMENT PORTAL

1. Introduction

Requirements Gathering is the second phase of the Student Internship and Placement Management Portal project. Industrial Assignment 1 defined the vision, scope, user roles and the seven epics with 28 tasks. This assignment identifies, collects, analyses and documents what the stakeholders actually need from the system, so that the user stories, database, user interface and code developed later are based on clear and agreed requirements.

2. Objectives of Requirements Gathering

- Understand how internships and placements are managed in the college at present and where the difficulties are.
- Identify the needs of every user role: Student, Company Recruiter, Placement Officer, Faculty Coordinator and Administrator.
- Document functional requirements, non-functional requirements and business rules in a clear and testable form.
- Prioritise the requirements so that the most important features are developed first.
- Create traceability from each epic and task to its requirements and user stories.

3. Requirement Gathering Techniques

Technique	Source	Information Collected
Interview	Placement Officer and faculty coordinators	Current placement process, approval rules, eligibility policy, reports prepared for management.
Interview	Company HR / visiting recruiters	Details needed to shortlist students, number of selection rounds, how offers are communicated.
Questionnaire / Survey	Final-year diploma students	How opportunities are discovered, problems faced in applying, features expected from a portal.
Observation	Training and Placement cell office	Use of registers, spreadsheets, notice boards and messaging groups; time spent on manual work.
Document Analysis	Previous placement lists, company notices, eligibility criteria sheets, report formats	Data fields to be stored; formats of existing records and reports.
Benchmarking	Existing college and job portals	Useful features (search, filters, status tracking) and common gaps.
Brainstorming	Project team and faculty guide	Feature ideas, role boundaries and priorities.

4. Stakeholder Interview and Survey Questions

The following questions were prepared for each stakeholder group. The answers were analysed and converted into the requirements listed in Section 7.

4.1 Placement Officer

Question	Requirement Identified
How are companies and their requirements recorded at present?	Central company profile and opportunity records (FR-15, FR-16).
Which eligibility rules do you apply to students?	Eligibility on CGPA, branch, semester and deadline checked by the system (FR-20, FR-21).
How do you approve and communicate with visiting companies?	Recruiter account approval by the Placement Officer (FR-08).
Which reports do you prepare for the Principal and for accreditation?	Branch-wise, company-wise and period-wise reports (FR-32, FR-33).

4.2 Students

Question	Requirement Identified
How do you get information about internships and jobs?	Single portal listing all open opportunities with search and filters (FR-34, FR-35).
What problems do you face while applying?	One profile and resume for all applications; online apply (FR-10, FR-14, FR-20).
How do you know whether you are shortlisted or selected?	Application status tracking and status history (FR-22, FR-24).
How are interview dates and offers communicated to you?	Interview schedule and offer response in the portal (FR-27, FR-29).

4.3 Company Recruiters

Question	Requirement Identified
What details of a candidate do you need before shortlisting?	Student profile with branch, CGPA, skills and resume visible to recruiters of applied candidates (FR-11).
How many selection rounds do you conduct?	Status values and interview scheduling for multiple rounds (FR-23, FR-26).

Question	Requirement Identified
How do you want to receive resumes and communicate results?	Applicant list with resume access; status update visible to students (FR-22, FR-23).

4.4 Faculty Coordinators and Administrator

Question	Requirement Identified
What department-level information do you need?	Read-only department summaries and reports (FR-32, FR-33).
Who should be able to create, change or remove user accounts?	Administrator-controlled access management with history (FR-07, FR-09).
What security expectations do you have?	Password hashing, role-based access, session timeout (FR-03, FR-06, NFR-02).

5. Analysis of the Existing System

5.1 Current Process

1. Companies inform the T&P cell about a drive by phone or e-mail.
2. The T&P cell announces the drive on the notice board or in messaging groups.
3. Students submit paper forms or send resumes by e-mail.
4. The T&P cell checks eligibility manually and prepares a list in a spreadsheet.
5. Shortlist, interview dates and results are announced informally.
6. Placement statistics are compiled manually at the end of the year.

5.2 Problems Identified

Problem	Effect
Information is scattered	Students miss opportunities; records are incomplete and duplicated.
Manual eligibility checking	Delay and human error; ineligible students may be considered.
No status tracking	Students and companies do not know the current stage of an application.
Informal communication	Interview dates and offers are missed or misunderstood.
Manual reporting	Reports are late and may be inaccurate.
No access control	Student data may be seen by people who do not need it.

6. Proposed System

The proposed portal replaces the manual process with an integrated workflow: Register → Create profile → Post opportunity → Apply (with automatic eligibility check) → Review and update status → Schedule interview → Release offer → Respond to offer → Dashboards and reports. Every step is controlled by role-based permissions and every important change is stored in a history table.

7. Functional Requirements

Priority follows the MoSCoW method: **Must** (essential), **Should** (important but not critical) and **Could** (desirable).

ID	Epic	Task	Requirement	Priority
FR-01	EP01	T01	The system shall allow users to register using valid information (name, e-mail, phone, role-specific details and password).	Must
FR-02	EP01	T01	The system shall reject registrations with duplicate e-mail, phone or enrollment number, empty mandatory fields or invalid formats and show the exact error.	Must
FR-03	EP01	T01	The system shall store passwords using one-way hashing and never display them in plain text.	Must
FR-04	EP01	T02	The system shall allow registered users to securely log in with e-mail/phone and password.	Must
FR-05	EP01	T02	The system shall lock an account temporarily after repeated failed login attempts and notify the user.	Should
FR-06	EP01	T03	The system shall allow logged-in users to log out and shall end idle sessions automatically.	Must
FR-07	EP01	T04	The system shall allow the Administrator to approve, reject, activate, deactivate and delete accounts, and to assign or change roles.	Must
FR-08	EP01	T04	The system shall allow the Placement Officer to approve or reject Company Recruiter accounts.	Must
FR-09	EP01	T04	The system shall record every access, role and status change in an access history.	Should
FR-10	EP02	T05	The system shall allow a student to create a profile with branch, semester, CGPA, skills, address and contact details.	Must
FR-11	EP02	T06	The system shall allow authorized users to view student profiles according to their role.	Must

ID	Epic	Task	Requirement	Priority
FR-12	EP02	T07	The system shall allow a student to edit the profile and shall record every change with old and new values.	Must
FR-13	EP02	T07	The system shall allow the Placement Officer to verify or correct academic details such as CGPA.	Should
FR-14	EP02	T08	The system shall allow a student to upload, replace, view and delete a resume in an accepted format and size.	Must
FR-15	EP03	T09	The system shall allow an approved recruiter to create and update the company profile.	Must
FR-16	EP03	T10	The system shall allow posting an internship or placement opportunity with title, description, location, eligibility, compensation and deadline.	Must
FR-17	EP03	T11	The system shall allow editing an opportunity and shall keep the edit history.	Should
FR-18	EP03	T12	The system shall allow closing an opportunity so that new applications are blocked.	Must
FR-19	EP03	T12	The system shall allow soft deletion of an opportunity with a reason and shall allow authorized restoration.	Should
FR-20	EP04	T13	The system shall allow an eligible student to apply to an open opportunity before its deadline.	Must
FR-21	EP04	T13	The system shall prevent duplicate and ineligible applications and explain the reason.	Must
FR-22	EP04	T14	The system shall allow students to view their own applications and authorized users to view applications within their scope.	Must
FR-23	EP04	T15	The system shall allow recruiters and the Placement Officer to update application status to Shortlisted, Selected or Rejected.	Must
FR-24	EP04	T15	The system shall save every status change with date, user and reason in a status history.	Must
FR-25	EP04	T16	The system shall allow a student to withdraw an application that is not yet finalized.	Should
FR-26	EP05	T17	The system shall allow scheduling an interview for a shortlisted application with date, time, mode and venue/link.	Must
FR-27	EP05	T18	The system shall allow students and authorized users to view interview schedules.	Must

ID	Epic	Task	Requirement	Priority
FR-28	EP05	T19	The system shall allow releasing an offer only for a selected application.	Must
FR-29	EP05	T20	The system shall allow a student to accept or reject a pending offer and shall record the decision.	Must
FR-30	EP06	T21	The system shall show a role-specific dashboard after login.	Must
FR-31	EP06	T22	The system shall show summaries of opportunities by type, status and company.	Should
FR-32	EP06	T23	The system shall show placement summaries such as applications, selections, offers and placed students by branch.	Should
FR-33	EP06	T24	The system shall generate reports for a selected period and scope and record each report generated.	Should
FR-34	EP07	T25	The system shall allow searching opportunities by title, company and keyword.	Must
FR-35	EP07	T26	The system shall allow filtering opportunities by type, status, location, eligibility and deadline.	Must
FR-36	EP07	T27	The system shall allow authorized users to search student records within their permitted scope.	Should
FR-37	EP07	T28	The system shall allow filtering applications by status, opportunity, company, branch and date.	Should

8. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Login, search and filter requests shall respond within 3 seconds for up to 500 concurrent users.
NFR-02	Security	Passwords shall be stored as hashes; sessions shall time out after inactivity; all inputs shall be validated; access shall be controlled by role.
NFR-03	Privacy	Student contact details and resumes shall be visible only to authorized roles; list views shall show only necessary fields.
NFR-04	Usability	The interface shall be simple, consistent and responsive on desktop and mobile browsers, with clear error messages.

ID	Category	Requirement
NFR-05	Reliability	The system shall be available for at least 99% of the time during the placement season; database backup shall be taken daily.
NFR-06	Accuracy	Summaries and reports shall always reflect the current stored records and statuses.
NFR-07	Auditability	Access changes, profile edits, opportunity edits and deletions, status changes and reports shall be recorded with user and time.
NFR-08	Scalability	The design shall support more students, companies and records without redesign.
NFR-09	Maintainability	The backend shall follow a layered structure (controller, service, repository) with documented APIs.
NFR-10	Compatibility	The portal shall work on current versions of Chrome, Edge and Firefox.

9. Business Rules

ID	Business Rule
BR-01	A user must be authenticated before accessing any protected feature.
BR-02	Invalid or incomplete data shall never be accepted by the system.
BR-03	Company Recruiter, Placement Officer and Faculty Coordinator accounts remain pending until approved; a pending account cannot use protected features.
BR-04	Only opportunities posted by an approved recruiter with a company profile are visible to students.
BR-05	A student can apply only to an open opportunity, before the deadline, if the student meets the minimum CGPA, branch and semester conditions.
BR-06	A student can apply to the same opportunity only once; a withdrawn application can be re-submitted only before the deadline.
BR-07	Every application status change must be saved in a history; earlier states must never be overwritten.
BR-08	An interview can be scheduled only for a shortlisted application and an offer can be released only for a selected application.
BR-09	A student who has accepted a placement offer cannot apply for further placement opportunities unless the Placement Officer grants an exception; internship applications are not affected.

ID	Business Rule
BR-10	Closed or deleted opportunities, profiles and accounts are kept in history and are not permanently erased by normal users.
BR-11	Users can access only the information permitted for their role.
BR-12	Reports and summaries must reflect the current status of records.

10. Role and Task Access Summary

The table summarises which roles may perform each epic's main activities. Detailed permission matrices for every task are given in the user stories (Industrial Assignments 3 and 4).

Epic	Guest	Student	Recruiter	Placement Officer	Faculty Coord.	Admin
EP01 User Management	Register	Own account	Own account	Approve companies	Own account	Full control
EP02 Student Profile	No	Own profile	View applicants	View, verify, edit	View (dept.)	View
EP03 Company and Opportunity	No	View open	Own company and posts	Manage all	View	Manage, restore
EP04 Application	No	Apply, view, withdraw	Review own posts	Review, override	View (read-only)	View
EP05 Interview and Offer	No	View, respond	Schedule, offer	Schedule, offer	View	View
EP06 Dashboard and Reports	No	Own dashboard	Own summary	All reports	Dept. reports	All reports
EP07 Search and Filter	No	Opportunities	Own posts, applicants	All records	Dept. records	All records

11. Data Requirements

The system must store the following main groups of data. Their detailed attributes and relationships are defined in Industrial Assignments 5 and 6.

Data Group	Main Information
Users and sessions	Name, e-mail, phone, password hash, role, status, role-specific details, login and logout times, session token.
Access history	Account, action type, old and new role or status, reason, action performed by, time.
Student profiles and resumes	Enrollment number, branch, semester, CGPA, skills, address, resume file details, verification status.
Profile edit history	Student, field changed, old value, new value, reason, edited by.
Companies	Company name, industry, website, location, HR contact, description.
Opportunities	Title, type (Internship / Placement), description, location, eligibility, compensation, deadline, status, posted by.
Opportunity edit and deletion history	Old and new values, reason, deleted by, restored time.
Applications and status history	Student, opportunity, applied date, status; old status, new status, changed by, reason, time.
Interviews	Application, date and time, mode, venue or link, round, result.
Offers	Application, offer date, package or stipend, status, response date.
Dashboard preferences and reports	Widget visibility and order; report type, period, scope, format, generated by, time.

12. Interface Requirements

- **User interface:** responsive web pages with role-based navigation, forms with validation messages, tables with search and filters, and status badges with text labels.
- **Software interface:** REST APIs exchanging JSON between the web application and the Spring Boot backend; database access through Spring Data JPA.
- **Communication interface:** on-screen notifications for status, interview and offer updates; e-mail notification can be added as a future enhancement.
- **Hardware interface:** any computer or mobile device with a modern web browser and internet connection.

13. Assumptions, Constraints and Dependencies

- **Assumption:** all users have a valid e-mail address and phone number and basic digital skills.
- **Assumption:** academic data entered by students is verified by the Placement Officer.

- **Constraint:** internet connectivity is required for all operations.
- **Constraint:** only web access to a single college is supported in the first version.
- **Dependency:** registration (T01) and login (T02) must be completed before any other task; applications (T13) depend on opportunities (T10) and profiles (T05); offers (T19) depend on selection (T15).

14. Requirement Prioritisation (MoSCoW)

Category	Requirements
Must have	FR-01 to FR-04, FR-06 to FR-08, FR-10 to FR-12, FR-14 to FR-16, FR-18, FR-20 to FR-24, FR-26 to FR-30, FR-34, FR-35
Should have	FR-05, FR-09, FR-13, FR-17, FR-19, FR-25, FR-31 to FR-33, FR-36, FR-37
Could have	E-mail notifications, saved filter presets, dashboard widget customisation
Won't have (this version)	Online tests, video interviews, mobile application, resume parsing, ERP integration

15. Requirements Traceability

Each epic and task is traced to its functional requirements and user story. Epic-local story numbers (EPxx-USyy) are used in the user-story documents.

Task	Task Name	Functional Requirements	User Story
T01	Register User	FR-01, FR-02, FR-03	EP01-US01
T02	User Login	FR-04, FR-05	EP01-US02
T03	User Logout	FR-06	EP01-US03
T04	Manage User Access	FR-07, FR-08, FR-09	EP01-US04
T05	Create Student Profile	FR-10	EP02-US01
T06	View Student Profile	FR-11	EP02-US02
T07	Edit Student Profile	FR-12, FR-13	EP02-US03
T08	Upload and Manage Resume	FR-14	EP02-US04
T09	Create Company Profile	FR-15	EP03-US01
T10	Post Opportunity	FR-16	EP03-US02
T11	Edit Opportunity	FR-17	EP03-US03
T12	Close or Delete Opportunity	FR-18, FR-19	EP03-US04
T13	Apply for Opportunity	FR-20, FR-21	EP04-US01

Task	Task Name	Functional Requirements	User Story
T14	View Application Details	FR-22	EP04-US02
T15	Update Application Status	FR-23, FR-24	EP04-US03
T16	Withdraw Application	FR-25	EP04-US04
T17	Schedule Interview	FR-26	EP05-US01
T18	View Interview Schedule	FR-27	EP05-US02
T19	Release Offer	FR-28	EP05-US03
T20	Respond to Offer	FR-29	EP05-US04
T21	View Dashboard	FR-30	EP06-US01
T22	View Opportunity Summary	FR-31	EP06-US02
T23	View Placement Summary	FR-32	EP06-US03
T24	Generate Reports	FR-33	EP06-US04
T25	Search Opportunities	FR-34	EP07-US01
T26	Filter Opportunities	FR-35	EP07-US02
T27	Search Student Records	FR-36	EP07-US03
T28	Filter Applications	FR-37	EP07-US04

16. Validation and Sign-off

- Requirements were reviewed with the faculty guide and the Placement Officer for correctness and completeness.
- Each requirement is clear, testable, traceable to a task and has a priority.
- Changes to requirements will be recorded and applied to the user stories, database, design and code.
- Approval of the requirements allows the project to move to the User Stories phase.

17. Conclusion

Requirements were gathered through interviews, surveys, observation, document analysis and benchmarking, and documented as 37 functional requirements, 10 non-functional requirements and 12 business rules. They are traced to the 28 tasks of the seven epics and will be converted into detailed user stories in Industrial Assignments 3 and 4.